

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

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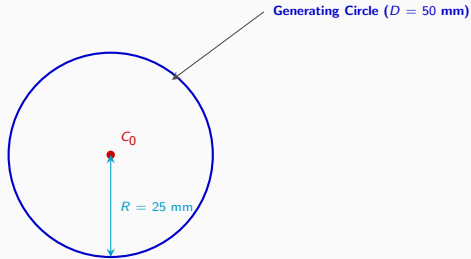
Problem Statement

Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

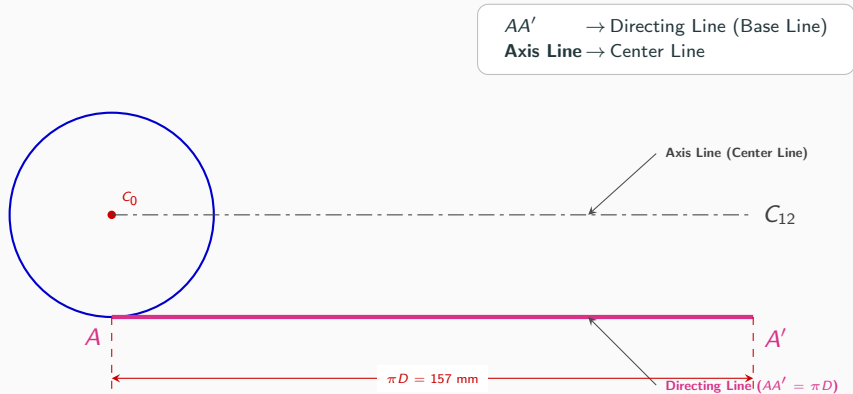
Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw the Generating Circle ($D = 50 \text{ mm}$, $R = 25 \text{ mm}$)



C_0 → Initial Center Point
Circle → Generating Circle

Step 2: Draw Directing Line AA' ($\pi D = 157 \text{ mm}$) and Axis Line

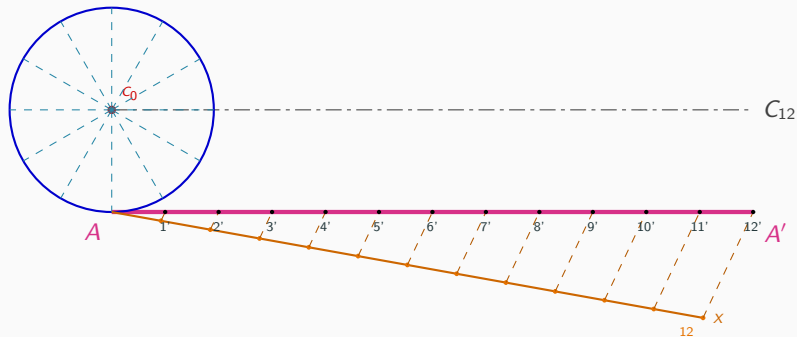


Step 3: Divide Generating Circle into 12 Equal Parts

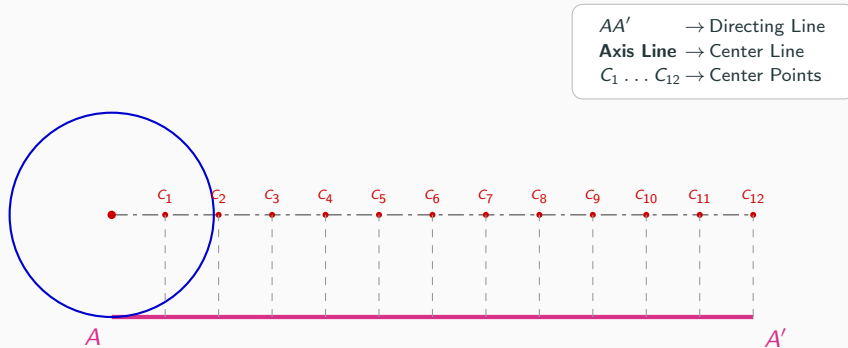


Step 4: Divide Directing Line AA' using Line Division Method

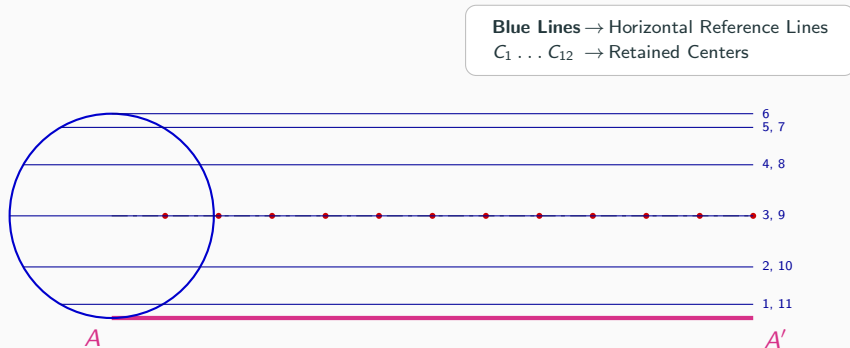
AA' → Directing Line
 AX → Inclined Line
 $1' \dots 12'$ → Parallel Divisions



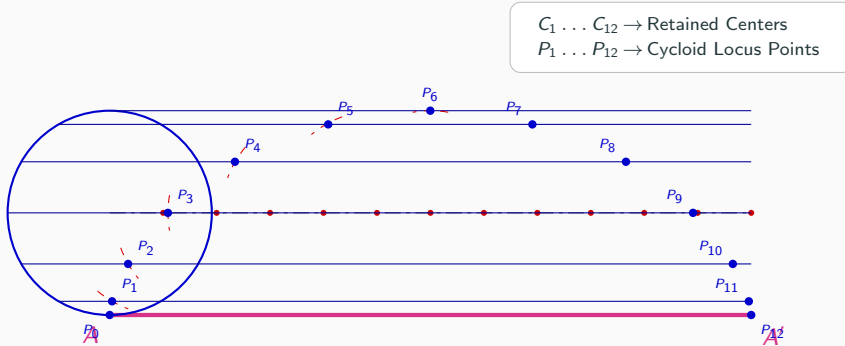
Step 5: Project Divisions to Locate Centers $C_1 \dots C_{12}$



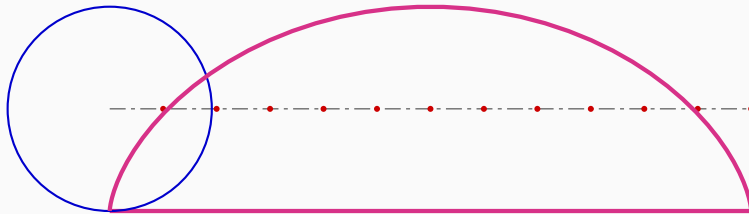
Step 6: Draw Horizontal Reference Lines



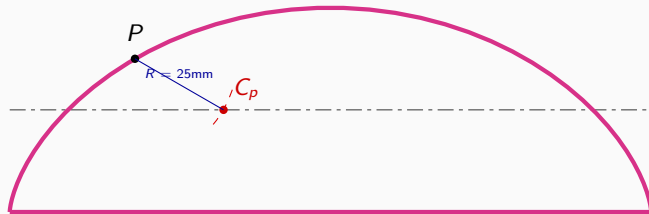
Step 7: Strike Arcs ($R = 25$ mm) to Locate Points $P_1 \dots P_{12}$



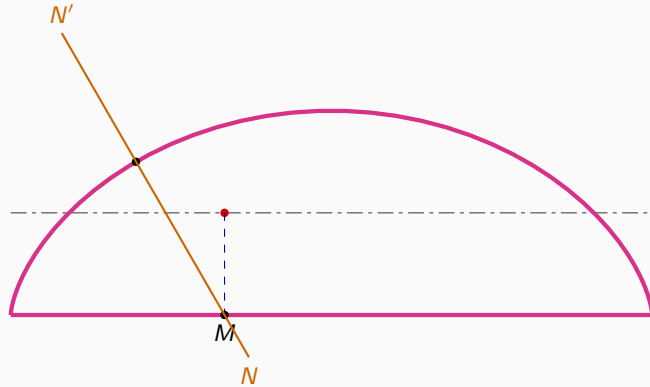
Step 8: Smoothly Join Points $P_0 \dots P_{12}$



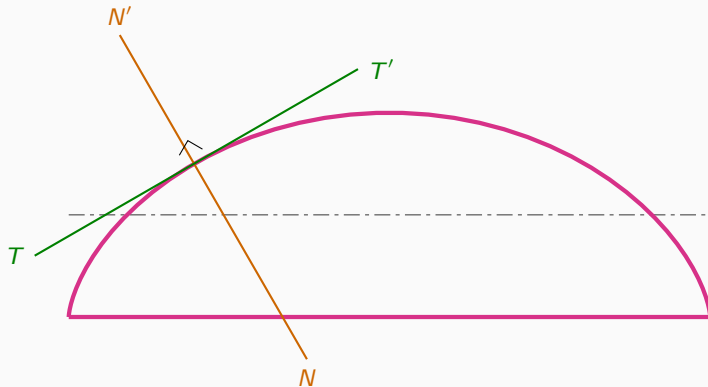
Step 9: Mark Point P & Strike Arc to Axis Line



Step 10: Project C_p Vertically to Locate Normal $N-N'$



Step 11: Draw Tangent Line $T-T'$ Perpendicular to Normal



Step 12: Final Completed Cycloid with Tangent & Normal

